



CPS INSTITUTE

LMS

SomaPRO - Learning Management System

PRODUCT OVERVIEW

The document contains the Product overview, its contents, tasks and capabilities. This is not a user Guide.

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SomaPRO Learning Management System - User Journey & Process Flow

Overview

SomaPRO is a comprehensive Learning Management System (LMS) that facilitates online education through a structured workflow involving multiple user roles: Students, Teachers, Administrators, and Managers.

Pre-LMS process (Eligibility);

Student

Pays **Registration** fee manually to get eligible to be registered by admin. Once registered they are issued with SomaPRO login credentials

Teacher

Goes through a recruitment process then pays registration fee to become eligible to be registered into SomaPRO. Once registered they are issued with SomaPRO login credentials

1. User Registration & Authentication

Registration Process

- **New User Registration:** System Administrator registers the eligible users above (teachers and students)
- **Email Verification:** System sends verification link to confirm email address (not implemented yet)
- **Initial Role Assignment:** Admin assigns roles (Student or Teacher)

Login Process

- Users log in using email and password
- **Role-based** dashboard redirection occurs automatically
- Session management with token storage in localStorage (i.e. system can always auto-log you in and you continue from where you left)

Role-Based Access Control

- **Admin:** Full system access, user management, course management, financial oversight
 - **Manager:** Limited administrative functions, user management, course management
 - **Teacher:** Course creation, course content management, grading.
 - **Student:** Course enrollment, learning, quizzes & assignments, grades
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2. Role Assignment & Management

Admin Functions

- **Create/Edit Roles:** Assign roles to new users
- **Role Modification:** Change user roles as needed
- **Permission Management:** Set access levels for different roles
- **Teacher Assignment:** Approve teacher applications and assign courses
- **Student Management:** Review and approve student registrations
- **Course Approvals:** Review and approve course publications
- **Course Management:** Create Course, add course content, grading
- **Enrollment Management:** Process student enrollments

Manager Functions

- **User Oversight:** Monitor user activities
 - **Course Approvals:** Review and approve course publications
 - **Enrollment Management:** Process student enrollments
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3. Course Creation & Content Development

Course Setup (Teacher/Admin)

1. Create Course

- Add course title, short name, description
- Set category, price (free/paid)
- Define start and end dates
- Set visibility (Public/Hidden) and status (Draft/Published)

2. Add Course Sections

- Organize content into logical sections
- Add section titles and descriptions
- Reorder sections using drag-and-drop

3. Create Lessons

- **Text Lessons:** HTML-formatted content with multimedia
- **Video Lessons:** Embed YouTube/Vimeo links or upload videos
- **Quiz Lessons:** Create assessments with multiple questions
- **Assignment Lessons:** Set tasks with due dates

4. Quiz Configuration

- Add multiple choice questions
- Set correct answers and points
- Define passing score (default 70%)
- Option to randomize questions

5. Assignment Setup

- Set assignment title and description
- Define due date (cannot be past date)
- Set maximum score (default 100)
- Attach resources or instructions

6. Course Publishing

- Preview course content
 - Submit for approval (if required)
 - Publish to course catalog
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4. Student Enrollment & Payment Process

Course Discovery

- **Browse Courses:** Students view available courses in catalog
- **Search & Filter:** Find courses by category, instructor, or price
- **Course Preview:** View course details, syllabus, instructor info

Enrollment Process

Step 1: Course Selection

- Student selects a course from catalog
- Views course details, price, and requirements

Step 2: Application Submission

- Click "Enroll Now" button
- System creates application record with status "PENDING"
- For paid courses, payment process initiates

Step 3: Payment Processing

- **Free Courses:** Direct approval after submission
- **Paid Courses:**
 - Student enters payment details
 - System processes payment (MPESA/Card/Bank Slip/Receipt No.)
 - Payment status updates to "COMPLETED"
 - Application status changes to "PAID"

Step 4: Admin Approval

- Admin reviews pending applications
- For paid courses: Verifies payment completion
- Approves or declines enrollment
- Status updates to "APPROVED" or "DECLINED"

Step 5: Enrollment Confirmation

- Student receives notification
- Course appears in "My Courses" list
- Access to course content granted
- Enrollment tracked in system

Payment Tracking

- **Revenue Dashboard:** Admin views total revenue
 - **Transaction History:** All payments recorded
 - **Monthly Reports:** Revenue trends over time
 - **Payment Status:** Track paid, pending, or failed transactions
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5. Learning Experience

Course Access

- **My Courses Dashboard:** Students see enrolled courses
- **Course Progress:** Visual progress indicators
- **Resume Learning:** Continue from last viewed lesson

Lesson Navigation

- **Structured Layout:** Sections with expandable lesson lists
- **Sequential Learning:** Mandatory lessons must be completed in order
- **Progress Tracking:** System marks completed lessons

Quiz Experience

- **Take Quiz:** Access quiz lessons from course content
- **Question Navigation:** Move between questions
- **Timed Attempts:** Track time spent (if configured)
- **Automatic Grading:** System calculates score instantly
- **Feedback:** Show correct answers and explanations
- **Pass/Fail:** Indicate if passing score achieved

- **Retake Option:** Allow students to retake (if configured)

Assignment Experience

- **View Assignment:** Read instructions and requirements
- **File Upload:** Submit documents, images, or code
- **Text Submission:** Provide written answers
- **Due Date Tracking:** Show remaining time
- **Submission Status:** Track submitted, pending, or graded
- **Late Submission Handling:** Apply penalties if configured

Progress Tracking

- **Lesson Completion:** Checkmark on completed items
 - **Course Progress Bar:** Visual percentage completion
 - **Grade Tracking:** View grades for graded items
 - **Activity Log:** Track last access and activity
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6. Grading & Assessment

Teacher Grading Process

1. **Access Grading Dashboard**
 - View all pending submissions
 - Filter by course or assignment
2. **Review Submissions**
 - Open student assignment
 - View submitted content or files
 - Provide feedback comments
3. **Assign Grade**
 - Enter grade (0-100 or custom range)
 - Add constructive feedback
 - Option to request resubmission

4. **Grade Publication**

- Grades automatically visible to students
- Student receives notification
- Grade appears in gradebook

Quiz Grading

- **Automatic:** Multiple-choice questions graded instantly
- **Manual:** Essay questions require teacher review
- **Combined Scoring:** Mix of auto and manual grading

Gradebook Features

- **Student View:** See all course grades in one place
 - **Letter Grades:** Convert percentages to letter grades (A, B, C, etc.)
 - **Grade Distribution:**
 - Bar charts showing grade ranges
 - Average scores per assignment
 - Class performance analytics
 - **Transcript Generation:** Export grade reports
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7. Reports & Analytics

Student Reports

- **Progress Report:** Track completion percentage across courses
- **Grade Report:** View all grades by course and assignment
- **Assignment Report:** List of completed and pending assignments
- **Time Tracking:** Hours spent on learning activities

Teacher Reports

- **Course Analytics:** Student engagement metrics

- **Student Performance:** Individual student progress tracking
- **Assignment Analysis:** Submission rates and grade distribution
- **Grade Distribution:** Statistical breakdown of grades
- **Class Summary:** Overview of all students in course

Admin Reports

- **Financial Reports:**
 - Total revenue from paid courses
 - Monthly revenue trends
 - Transaction history
 - Pending payments
- **System Analytics:**
 - User registration trends
 - Course enrollment statistics
 - Completion rates
 - Active users metrics
- **Performance Metrics:**
 - Average grades across platform
 - Most popular courses
 - Teacher performance indicators

Report Features

- **Download Options:** Export as PDF, CSV, or Excel
 - **Date Filtering:** View reports by date range
 - **Chart Visualizations:** Graphs and pie charts
 - **Real-time Data:** Live updates from database
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8. Key System Notifications

Automated Notifications

- **Enrollment Confirmation:** When application approved
- **Payment Confirmation:** Receipt after successful payment
- **New Assignment:** Alert when teacher posts assignment
- **Submission Confirmation:** When student submits work
- **Grade Posted:** Notification when graded
- **Due Date Reminders:** Before assignment deadlines
- **Course Updates:** When new content added

Notification Channels

- **In-app Notifications:** Bell icon with dropdown
- **Email Alerts:** For important actions
- **Dashboard Widgets:** Summary of pending items